Department of Computer Science and Engineering
Academic Year: 2025 - 2026 (Odd Semester)

Name of the Faculty member	**: Mrs.S.Shunmuga Priya, AP/CSE**
Degree, Semester & Branch, Sec	**: B.E, V & CSE – 'B'**
Course Code & Title	**: CCS375 WEB TECHNOLOGIES**
Date & Time	**: 22.08.2025 & 3 PM to 3.25 PM**

Innovative Practice I - Description

- **Unit / Topic:** Unit I / HTML5 Elements
- **Course Outcome:** CO1
- **Topic Learning Outcome:** TLO1b
- **Activity Chosen:** Game Based Learning
- **Justification**
  HTML5 is the standard for creating modern, responsive, and interactive websites, replacing older HTML versions. In this game-based learning activity, students create an interactive game using HTML5 elements. Students learned the use of the <audio> tag, custom controls, and HTML5 elements through hands-on practice. This approach promoted teamwork, creativity, and a deeper understanding of web technologies and made the session interactive and engaging.

Time Allotted for the Activity: 25 Minutes

- **Details of Implementation**
  - In this game-based learning activity, students create an interactive game using HTML5 elements.
  - The webpage uses <section> to divide game stages, <article> for each question, and <button> for answer options. A <progress> element tracks the player's score or level. The <audio> tag plays a sound for correct or wrong answers, and the <video> tag can display a short tutorial after the game ends.
  - Students were divided into two teams to perform the activity.
  - Team A was responsible for capturing and playing bird sounds for the game-based learning activity. The team recorded and collected various bird sounds in formats like .mp3 or .wav and organized them for easy access.
  - Team B was responsible for listening to the bird sounds played by Team A and responding with the correct music with the matching clip and then explains the concept in front of the class which is shown in Figure2. Groups of students worked together where one group handled music playback, another managed answer playback, and others explained the working of their players.

- Using the HTML5 <audio> element, each sound was embedded into the webpage, and custom controls were added with <button> elements for play and pause functionality. A progress bar was implemented using <input type="range"> to track playback, and volume sliders allowed adjustment of sound levels. Each sound was labeled using the <label> element and grouped within <div> or <section> elements to create a structured and visually organized interface.

CO – PO / PSO mapping:

CO →	**PO1**	**PO2**	**PO3**	**PO9**	**PO10**	**PO11**	**PSO1**
CO1	3	3	3	1	2	2	3

PO-PSO Mapped:

Innovative Practice	**PO1**	**PO2**	**PO3**	**PO9**	**PO10**	**PO11**	**PSO1**
	3	3	3	1	2	2	3
Justification for Correlation	Students apply core knowledge of web technologies (HTML5 <audio> elements, attributes) to implement an audio player.	Students analyze the requirements and solve challenges in integrating audio controls.	The task requires designing a functional UI and integrating features to create a working audio player.	The activity can be done collaboratively, promoting teamwork and sharing of creative ideas.	Students explain the working of HTML5 elements through presentations, and demonstrations.	Students plan tasks (design, coding, testing) in a time-bound manner, gaining exposure to project management concepts.	Students gain practical skills in applying web technologies (HTML5 audio APIs) for real-world applications.

- Images / Screenshot of the practice:

Figure 1

Figure 2

Reflective Critique

Outcome

- Students gained a deeper understanding of HTML5 multimedia elements such as `<audio>`, `<video>`, `<section>`, `<article>`, `<progress>`, and `<button>` through hands-on implementation rather than passive learning. Developed problem-solving skills while debugging synchronization and responsiveness issues.
- The game-based setup transformed a theoretical topic into an enjoyable, interactive session, improving students' participation and motivation.
- Enhanced confidence to design mini real-world applications in web development and apply theoretical knowledge in practice.

Feedback of Practice from Students:

- Students found the activity engaging and enjoyable once they understood the concepts.
- Students were able to apply theoretical concepts to create a practical web component.

Benefits of the Practice:

- Students gained both theory and practical exposure to HTML5 elements.
- They learned to implement interactive features such as play/pause, progress bar, and volume slider.
- Games capture students' interest and make them more motivated to learn.
- Helps students to tackle web technology questions effectively in exams.

Challenges Faced in Implementation :

- A few students found it difficult to synchronize the progress bar with audio playback.
- Debugging CSS issues was initially confusing for beginners.

References

https://helpfulprofessor.com/digital-game-based-learning/
https://study.com/academy/lesson/game-based-learning-definition-and-examples/